Design and Analysis of Algorithm (CS3004-1)
Multiple Choice Questions
Unit-1

1` Main objective of analysing the algorithm is

a) To design algorithms
b)To implement algorithms
c) To evaluate for efficiency
d) To debug algorithms

2 Which of the following is NOT a characteristic of a good algorithm?

a) Clarity and simplicity
b) Efficiency in terms of time and space
c) Correctness
d) The use of complex data structures

3 What does "optimization" refer to in the context of algorithms?

a) The process of making an algorithm more complex
b) The process of improving an algorithm's efficiency
c) The process of making an algorithm run slower
d) The process of making an algorithm more clear

4 What is the purpose of pseudocode in algorithm development?

a) To provide a step-by-step implementation guide
b) To prevent others from understanding the
c) To hide the algorithm's logic
d) To succinct the algorithm

5 Characteristics of an algorithm are

a) Input, output, definite, finite, connective.
b) input, output, definite, finite, effective.
c) input, output, reflective, finite, effective
d) input, output, reflective, finite, connective.

6 When an algorithm is written in the form of a programming language, it becomes a _

a) Flowchart
b) Program
c) Pseudo code
d) Syntax

7 Pseudocode is written using

a)Combination of Native & programming language
b) High level Programming language only
c) Natural language only
d) Low level programming language

8 Which of the following is incorrect?
Algorithms can be represented:

a)as pseudo codes
b)as syntax
c) as programs
d) as flowcharts

9 What will be the output of the following pseudocode if the driver is fun(5)?

```
function fun(n)
        if n == 1 return 1 else return n + fun(n-1)
```

a)120
b)24
c)6
d)15

10 What will be the output of the following pseudocode if the driver is fun(18,24)

```
function fun(a,b)
      while(b!=0)
            fun(b, a mod b)
```

return a

a)18 b)24

c) 6 d)2

11` Given the following time complexities, which one is the slowest growing function as $n \to \infty$

a)$\mathbf{O(logn)}$ b)$O(n)$

c)$O(n^2)$ d)$O(2^n)$

12 Which of the following best describes the Big-Ω notation?

a) Describes the lower bound of the time complexity of an algorithm

b) Describes the upper bound of the time complexity of an algorithm

c) Describes the exact time complexity of an algorithm

d) Describes the average-case time complexity of an algorithm

13 What does the Big-O notation, **O(f(n))**, represent?

a) The average-case time complexity of an algorithm

b) The worst-case time complexity of an algorithm

c) The best-case time complexity of an algorithm

d) The exact time taken by the algorithm

14 The time complexity of an algorithm is $O(n^2)$

Which of the following statements is correct about this algorithm?

a) The algorithm's running time grows linearly with respect to the size of the input

b) The algorithm's running time grows quadratically with respect to the size of the input

c) The algorithm's running time grows logarithmically with respect to the size of the input

d) The algorithm's running time is constant irrespective of the input size

15 If an algorithm's time complexity is $O(2^n)$, what can you infer about the algorithm?

a) The algorithm will be efficient for large inputs

b) The algorithm's time complexity grows exponentially with the input size

c) The algorithm will run in constant time for all inputs

d) The algorithm's runtime will decrease as the input size increases

16 In asymptotic analysis, what does it mean if a function is said to be $O(n)$?

a) The time complexity increases exponentially as n increases

b) The algorithm's runtime grows linearly with the input size

c) The runtime of the algorithm is independent of the input size

d) The algorithm's runtime grows logarithmically as n increases

17 If an algorithm has a worst-case time complexity of $\mathbf{O(n^3)}$, which of the following time complexities would indicate a faster algorithm for large **n?**

a)$\mathbf{O(n^2)}$ b)$O(n^3)$

c)$O(n^4)$ d)$O(2^n)$

18 Consider an algorithm with the following time complexities for different scenarios:

- Best Case: $O(n)$
- Worst Case: $O(n^2)$

- Average Case: $O(n\log n)$

In which case will the algorithm perform the fastest?

a) Best Case b) Worst Case

c)Average Case d) All cases will perform equally fast

19 Which of the following is the correct asymptotic notation for expressing the upper bound of an algorithm's running time?

a)$\Omega(n)$ b)$\Theta(n)$

c)$\mathbf{O}(\mathbf{n})$ d)$o(n)$

20 If an algorithm has a time complexity of $\mathbf{O}(\mathbf{n}^3)$, which of the following functions represents its growth rate?

a) Linear **b) Cubic**

c) Exponential d) Logarithmic

21` What is the first step in the general plan for analyzing the time efficiency of recursive algorithms?

a) Identify the basic operation **b)Decide on a parameter indicating input size**

c) Set up a recurrence relation d)Check whether the number of times the basic operation is executed can vary on different inputs

22 Which is the fundamental operation that is considered while analysing the algorithms?

a) Count of basic operation **b)** Arithmetic operation

c) Recurrence Equation **d)** Geometric operation

23 What is the goal of the Tower of Hanoi problem?

a) To move all the disks to the second peg **b)** To move all the disks to the first peg

c) To move all the disks to the third peg **d)** To move all the disks to the fourth peg

24 The time complexity of the solution tower of hanoi problem using recursion is

a) $O(n^2)$ **b) $\mathbf{O(2^n)}$**

c) O(n log n) **d)** O(n)

25 **What is the typical time complexity of a non-recursive algorithm that processes each element in an array of size n exactly once?**

a) O(1) **b)** O(n log n)

c) O(n) **d)** O(n^2)

26 **Which of the following statements about brute force algorithms is true?**

a) They always provide the optimal solution but are inefficient for large problem sizes. **b)** They are the most efficient algorithms for all types of problems

c)They exhaustively search all possibilities, making them simple but often inefficient for large inputs **d)** They reduce the problem size in each iteration, unlike recursive algorithms

27 **What is the primary characteristic of a brute force algorithm?**

a) It uses a divide-and-conquer strategy to break problems into smaller subproblems.

b) It attempts to solve the problem by checking all possible solutions until the correct one is found.

c) It uses dynamic programming to avoid redundant calculations

d) It uses a greedy strategy to find an optimal solution.

28 What is the number of times the comparison operation is executed in the algorithm for finding the largest element in a list of numbers?

a) C(n) = n

b) C(n) = n – 1

c) C(n) = n + 1

d) C(n) = n - 1

29 What is considered as the basic operation in the algorithm for finding the largest element in an array of integers?

a) The assignment operation

b)The comparison operation

c) The addition operation

d) The multiplication operation

30 Which of the following recursive algorithms has a time complexity of O(n)?

a) Fibonacci sequence using naive recursion.

b) Merge sort.

c) Binary search on a sorted array.

d) Factorial computation using recursion.

31` The Selection Sort algorithm finds the lowest value in an array and moves it to the ___ of the array.

a) end

b) **front**

c) middle

d) particular location

32 The Bubble Sort algorithm finds the lowest value in an array and moves it to the ___ of the array.

a) end

b) front

c) middle

d) particular location

33 Linear search algorithm, also known as_____

a)Binary search

b) simple search

c)sequential search

d)Difficult search

34 **Exhaustive Search algorithm checks every possible solution to find the ____ one**

a) Worst

b) Average

c) amortized

d) Best

35 **Exhaustive search falls under which algorithm technique**

a) Brute force

b) Decrease and Conquer

c) Divide and Conquer

d) Greedy

36 A string-searching algorithm, sometimes called string-matching algorithm, is an algorithm that searches a body of text for portions that match by_____.

a) pattern

b) word

c) index

d) text

37 The time complexity of the naive string matching algorithm -m+1)

a) O(nm)

b)O(n+m)

c)O(n-m+1)

d)O(n)

38 The time complexity of the linear Search algorith

a) O(nm)

b)O(n+m)

c)O(n-m+1) **d)O(n)**

39 The time complexity of the Bubble sort algorithm

a) O(nm) **b)O(n^2)**

c)O(n-m+1) d)O(n)

40 The time complexity of the Selection sort algorithm

a) O(nm) b)**O(n^2)**

c)O(n-m+1) d)O(n)

41` What is the primary characteristic of the Divide and Conquer approach?

a) Solving problems by iteration

b) Solving problems by dividing them into overlapping sub problems

c) Dividing a problem into smaller independent subproblems, solving them recursively, and combining their results

d) Dividing a problem into smaller problems and linearly solving it.

42 In Quick Sort, what is the worst-case time complexity?

a) O(n^2) **b)** O(n logn)

c) O(logn) **d)** O(n)

43 Binary Search works only on which type of arrays?

a) Unsorted arrays **b) Sorted arrays**

c) Arrays with distinct elements **d)** Arrays of integers

44 Which of the following algorithms is **NOT** based on the Divide and Conquer approach?

a) Merge Sort **b)** Quick Sort

c) Binary Search **d) Selection sort**

45 Which of the following is true about Merge Sort?

a) It is an in-place sorting algorithm

b) It has a worst-case time complexity of O(n^2)

c) It uses additional memory for merging subarrays

d) It always selects the first element as a pivot

46 During the merging process of Merge Sort, what is the time complexity to merge two sorted subarrays?

a) O(n) **b)** O(logn)

c) O(n^2) **d)** O(nlogn)

47 What is the average-case time complexity of merge sort?

a) Θ (logn) **b)** Θ (n)

c) Θ (n^2) **d) Θ(nlogn)**

48 How many sub arrays does the quick sort algorithm divide the entire array into?

a) one **b) two**

c) three **d)** four

49 What is the recurrence relation for merge sort algorithm?

a) C(n)=2C(n)+1 **b)** C(n)=2C(n/2)-1

c) **C(n)=2C(n/2)+n** **d)** C(n)=2C(n^2/2)+n

50 **What is the time complexity of Binary Search in the best case?**

a) Θ (n) **b)** Θ (logn)

c) Θ (nlogn) **d) $\boldsymbol{\Theta}$ (1)**